Determination of Aquifer Parameters from Step Tests and Intermittent Pumping Data

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ABSTRACT

The expression

$$s/Q_n = \frac{264}{T} \log_{10} \left[\frac{0.3\,T}{r^2 S} \beta_n(t)\,(t - \tau_n) \right]$$

is a handy tool for predicting aquifer parameters using all the drawdown and recovery data obtained during step-drawdown tests or periods of cyclic or intermittent pumping.

For inefficient wells the expression

$$s/Q_n = \frac{264}{T} \log_{10} \left[\frac{0.3\,T}{r^2 S} \beta_n(t)\,(t - \tau_n) \right] + CQ_n^y$$

fits pumping wells better. The data from conventional step-drawdown tests of pumping wells can be used to obtain T, C, and y. If the data fail to fit the ideal, they may be used to interpret local hydrologic conditions.

BACKGROUND

Two problems that hydrologists face frequently are (1) existing wells cannot be taken from service long enough to conduct controlled long-term pumping tests or cost prohibits long-term constant rate tests, and (2) tests of new wells are rarely conducted at ideal constant rates. To cope with these problems we have developed a relatively simple analytic technique that lets us use step-drawdown data and drawdown data obtained from wells that pump intermittently at different discharge rates.

This brief note presents the technique we use to evaluate drawdown and recovery data obtained during step-pumping tests or during several episodes of intermittent pumping. First, we present the simple general equation that relates the usual variables (drawdown, transmissivity, storativity, etc.) to step-wise changes in discharge. (Intermittent pumping equates to a step with zero discharge.) Then we show how the equation provides a basis for determining transmissivity and storativity of the water-yielding rocks using nearly all the data simultaneously. Next we discuss how departures of the data during conventional step-drawdown tests from the ideal provide other useful hydrogeologic information. Finally we give some examples from our field experience.

THEORY

For an ideally confined system with radial flow to a well with constant discharge, Q, the drawdown, s, is given by Theis' nonequilibrium formula as

$$s = \frac{Q}{4\pi T} W(u)\ .$$

Since the differential form of the equation and its boundary conditions are linear, for a step change in pumping rate, ΔQ, at time τ, the drawdown at any time, $t > \tau$, is given by the principle of superposition as

$$s = \frac{1}{4\pi T} [Q\,W(u) + \Delta Q\,W(u')] \quad t > \tau \qquad (1)$$

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Discussion open until September 1, 1980.

where $W(u) = \int_u^\infty \frac{e^{-u}}{u} du$ is the well function;

$u = \frac{r^2 S}{4Tt}$ and $u' = \frac{r^2 S}{4T(t-\tau)}$ are dimensionless variables; r is the radial distance from the center of the pumping well to a point of observation in the flow-system; and S and T are the storage coefficient and transmissivity.

If the change in pumping rate at time τ is from Q to zero (i.e., $\Delta Q = -Q$), the well is then said to be recovering, and Equation (1) takes the form

$$s = \frac{Q}{4\pi T} [W(u) - W(u')] \quad t > \tau \qquad (2)$$

Jacob (1950) noted that for small values of u′ (i.e., large values of elapsed time), Equation (2) can be approximated by

$$s = \frac{Q}{4\pi T} \ln \left(\frac{t}{t-\tau}\right) \qquad (3)$$

If t is in days, Q is in gallons per minute, T is gallons per day per foot, and r is in feet, Equation (3) can be generalized for a series of pumping and recovery periods (Figure 1) as

$$s = \begin{cases} \frac{264Q_n}{T} \log_{10}[(\frac{t_1}{t_1'})^{Q_1/Q_n} \cdots (\frac{t_{n-1}}{t_{n-1}'})^{Q_{n-1}/Q_n} \frac{t_n}{t_n'}] & (4a) \\ \quad t_n' > 0 \text{ during recovery} & \\ \frac{264Q_n}{T} \log_{10}[\frac{0.3T}{r^2S} \cdot (\frac{t_1}{t_1'})^{Q_1/Q_n} \cdots (\frac{t_{n-1}}{t_{n-1}'})^{Q_{n-1}/Q_n} \cdot t_n] & \\ \quad t_n > 0 \text{ during pumpage} & \ldots\ldots (4b) \end{cases}$$

where $t_i = t - \tau_i$ is the elapsed time since the ith pumping rate started; $t_i' = t - \tau_i'$ is the elapsed time since the ith pumping rate stopped; and τ_i and τ_i' are the times ith pumping rate started and ended respectively.

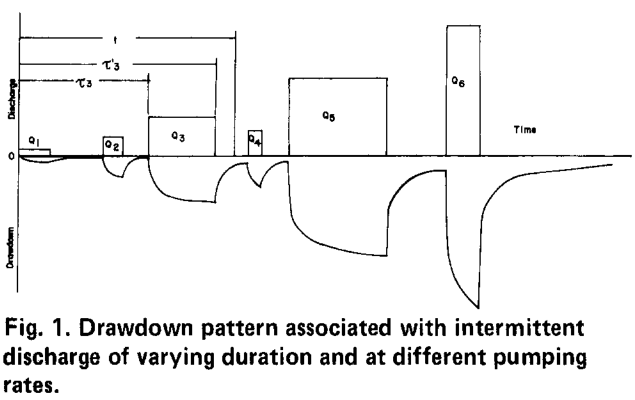

Fig. 1. Drawdown pattern associated with intermittent discharge of varying duration and at different pumping rates.

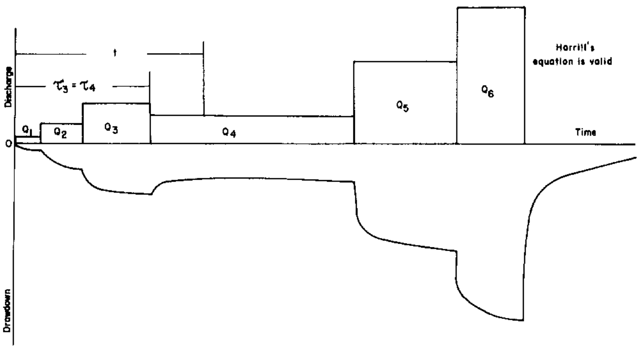

Fig. 2. Drawdown pattern caused by step pumping.

For the case where the pumping rate for several steps (Figure 2) changes but does not go to zero, the elapsed time since the end of the (i − 1)th pumpage is the same as that since the beginning of the ith pumpage (i.e., $t_{i-1}' = t_i$), then Equation (4a) reduces to

$$s = \frac{264Q_n}{T} \log_{10} \frac{t_1^{\Delta Q_1/Q_n} \ldots t_n^{\Delta Q_n/Q_n}}{t'} \quad \text{(Harrill, 1971)},$$

where t′ is the time since pumping stopped.

For the case where the intermittent pumping rate is constant (i.e., $Q = Q_1 = \ldots . Q_n$, Equation (4a) reduces to

$$s = \frac{264Q}{T} \log_{10} \left(\frac{t_1}{t_1'} \cdot \cdots \cdot \frac{t_n}{t_n'}\right) \quad \text{(Brown, 1963)}.$$

APPLICATION

Equations (4a) and (4b) are cumbersome to work with. However, if we let

$$\beta_n(t) = \begin{cases} \prod_{i=1}^{n-1} (\frac{t-\tau_i}{t-\tau_i})^{Q_i/Q_n} (\frac{t-\tau_1}{t-\tau_1'})^{Q_1/Q_n} \cdots (\frac{t-\tau_{n-1}}{t-\tau_{n-1}'})^{Q_{n-1}/Q_n} & n = 2, m \\ 1 & n = 1, \end{cases}$$

(where m is the total number of steps in a pumping test, and n is the number of the step we are interested in analyzing) and divide both sides of Equations (4a) and (4b) by Q_n, we get

$$\frac{s}{Q_n} = \begin{cases} \frac{264}{T} \log_{10}[\beta_n(t) \frac{t-\tau_n}{t-\tau_n'}] \quad t > \tau_n' \text{ during recovery} & (5a) \\ \frac{264}{T} \log_{10}[\frac{0.3T}{r^2S} \beta_n(t)(t-\tau_n)] \quad t > \tau_n \text{ during pumpage.} & (5b) \end{cases}$$

In these equations $\beta_n(t)$ can be thought of as an adjustment factor that takes previous pumpage into account. We define the expression

$$\beta_n(t)\ (t - \tau_n'),$$

as the *adjusted time* and the expression

$$\beta_n(t)\ \ (t - \tau_n)/(t - \tau_n'),$$

as *dimensionless time.* Appendix A is the program written for an HP-25 pocket calculator that we use to calculate $\beta_n(t)$ up to n = 4 for the cases where τ_{i-1} equals τ_i'.

Assuming that a pumping well is one hundred percent efficient, Equations (5a) and (5b) indicate that data plots on semilogarithmic graph paper of s/Q (n = 1, m) versus adjusted time for pumpage (or adjusted dimensionless time if recovery) should fall on the same straight line, which has a slope 264/T (Figure 3). This characteristic of Equations (5a) and (5b) allows hydrologists to make use of all data collected during step-pumping tests to calculate T and S.

$$T = \frac{264}{\Delta(s/Q)} \text{ one log cycle}$$

For an observation well (r feet from the pumping well) the intercept of the straight line drawn through the plotted points with s/Q = 0 gives t_o for the conventional determination of storativity

$$S = 2.08 \times 10^{-4}\ \ Tt_o/r^2$$

DISCUSSION

For evaluating well performance, transmissivity, and local hydrologic conductivity we have found that short step-drawdown tests are in many ways as useful as much longer constant-rate tests. Our typical step test consists of five thirty-minute steps and five hours of recovery. These tests rarely include observation wells.

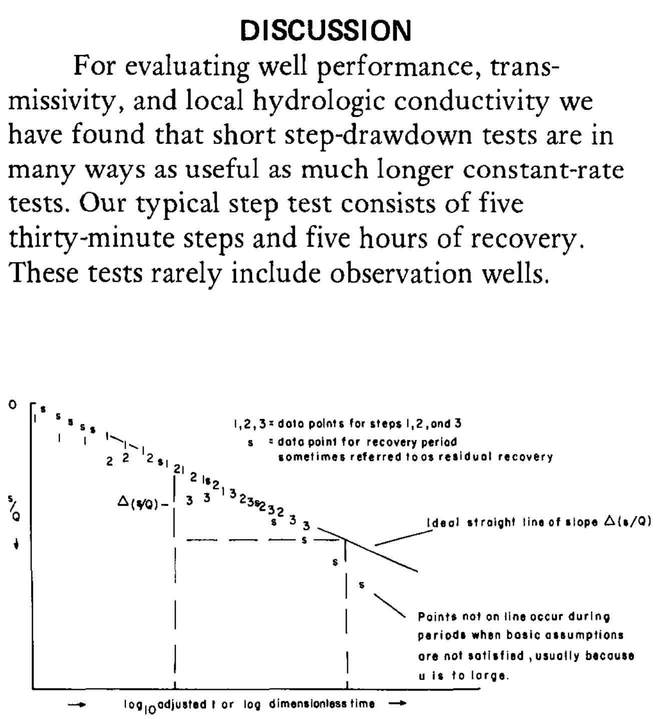

Fig. 3. Plot of drawdown versus log adjusted time and residual drawdown versus log dimensionless time in an ideal efficient pumping well.

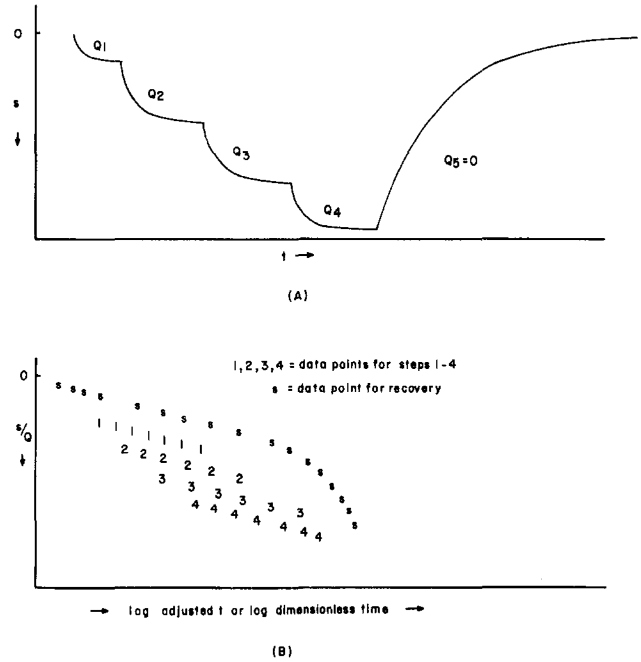

Fig. 4. (A) Drawdown pattern during conventional step test. (B) Pattern of drawdown versus adjusted or dimensionless time in inefficient pumping well.

Moreover, the data for the pumped well rarely conform to the ideal described above. In the following paragraphs we address the principal reasons for the lack of conformity. These reasons include:

(1) well losses,

(2) well is developing,

(3) hydrologic boundaries and local hydrologic conditions that violate the Theisian assumptions, and

(4) other factors.

Well Losses

Figure 4 illustrates a typical plot of data obtained at a pumped well during a step test. Instead of the data falling on one ideally straight line they lie on parallel lines and the early recovery data fell on a curve. These departures come about because the well itself imposes head losses that are not taken into account in Equations (5a) and (5b). These well losses may be calculated by determining the coefficients of the equation

$$s = BQ + CQ^z \quad \text{or} \quad s/Q = B + CQ^y \qquad (6a)$$

from the data of a step-drawdown test (Rorabaugh, 1953; Lennox, 1966; Sheahan, 1971).

Jacob (1950) argued that for most uses z = 2 was a sufficient approximation and he wrote Equation (6a) as

$$s/Q = B + CQ. \tag{6b}$$

To use this equation one only had to pick (s/Q, Q) at a given elapsed time for each step, plot these points in cartesian coordinates, and draw the best straight line through the points; B is the s/Q intercept and C is the slope of the line. Sheahan (1971) provided a type-curve technique for using the same data points to obtain B, C, and z.

With the following procedure we eliminate the arbitrary element of choosing ≡ z = 2 and picking one data set from each step: If we let the function $f(Q_n)$ represent a general well-loss function and add it to the right side of Equation (5b), we get

$$s/Q_n = \frac{264}{T} \ln \left[\frac{0.3T}{r^2S} \beta_n(t)(t - \tau_n) \right] + f(Q_n). \tag{7}$$

For a step change in Q and a given adjusted time we can write

$$\Delta\left(\frac{s}{Q_n}\right) = \Delta f, \tag{8a}$$

or

$$\frac{\Delta(s/Q_n)}{\Delta Q_n} = \frac{df}{dQ}, \tag{8b}$$

where $\Delta(s/Q_n)$ represents the parallel shift of the straight line for $\Delta Q_n (= Q_n - Q_{n-1})$ change in pumping rate (Figure 5A).

If we express the function $f(Q_n)$ as CQ^y, then we obtain

$$\frac{dCQ_n^y}{dQ_n} =$$

$$\frac{\Delta(s/Q_n)}{\Delta Q_n} = yCQ_n^{y-1}, \tag{9a}$$

or

$$\log_{10}\left[\frac{\Delta(s/Q_n)}{\Delta Q_n}\right] = (y-1)\log_{10} Q_n + \log_{10} yC, \tag{9b}$$

Plotting $[\Delta(s/Q_n)/(\Delta Q_n)]$ versus Q_n on log-log paper produces a straight line (Figure 5B) with slope of $y - 1$ and an intercept at $\log_{10} Q_n = 1$ of yC. That is

$$y - 1 = \Delta \log_{10}{}^{[\Delta(s/Q_n)/\Delta Q_n]}, \tag{10}$$

and

$$yC = \left.\frac{\Delta(s/Q_n)}{\Delta Q_n}\right|_{Q_n=1}. \tag{11}$$

If we assume y = 1 [z = 2], then

$$C = \frac{\Delta(s/Q_n)}{\Delta Q_n}. \tag{12}$$

Since the discharge Q involves stepwise changes, the expressions (8b)-(12) may not be strictly correct equations. They are introduced here only for convenience in estimating the unknown value of y. Starting from Equation (8a), an alternate procedure to estimate C and y is as follows:

From Equation (8a), we can write

$$\Delta_i = C(Q_{i-1}^y) \qquad i = 2, m$$

where $\Delta_i = (s/Q_i) - (s/Q_{i-1})$ is the difference in s/Q between two steps for any given adjusted time. This expression gives m − 1 nonlinear equations with two unknowns.

If y is assumed to be known beforehand, the average C can be calculated from

$$C = \frac{\sum_{i=2}^{m} \Delta_i}{Q_m^y - Q_1^y}. \tag{13}$$

Transmissivity can be obtained from the slope of the parallel lines, using

$$T = \frac{264}{\Delta_{(s/Q)}} \cdot \text{one log cycle}$$

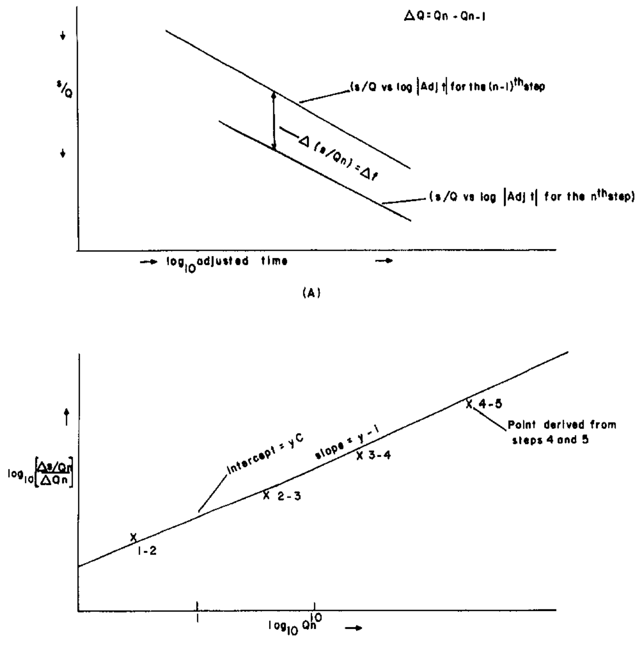

Fig. 5. (A) Illustration of Δ(s/Q). (B) Representative plot of data points used to determine well-loss coefficients.

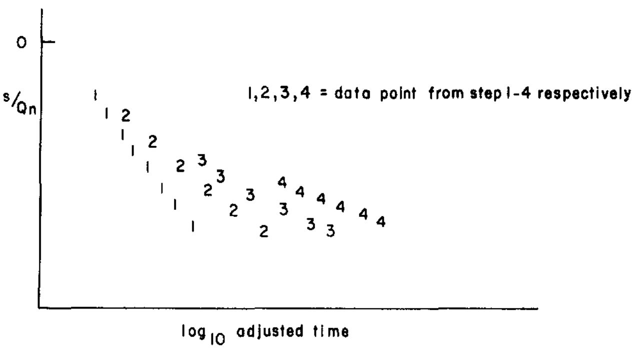

Fig. 6. Pattern of adjusted time versus s/Q_n data for developing pumping well.

Developing Well

If a well is developing, then the assumptions that the parameters C and y are constant are not valid. The result is a data suite that looks like that in Figure 6, which produce negative variable values of $\Delta(s/Q_n)$. Pumping well data from these tests cannot be used to estimate C, z, or T. Observation-well data are not affected by the well-loss factors of the pumping well, so T and S obtained in observation wells are valid. Tests giving data of this sort should be rerun after the well development is complete.

Hydrologic Boundaries and Other Factors

Equation (7) implies that the straight lines for increasing values of Q_n (n − 1, m) should be shifted parallel to each other in the direction of increasing s/Q_n.

Figure 7A illustrates a variation in the pattern we have seen in some test data. The lines are not straight as predicted from the theory, so Equation (7) may not represent adequately the complex nature of the real flow system. In this case an impermeable (negative) boundary condition may exist nearby or the thickness and hydraulic conductivity of the aquifer may vary with distance from the pumping well. Partial penetration and, in unconfined aquifers, large drawdown relative to the initial saturated thickness can also cause this kind of deviation. Figure 7B illustrates the pattern that constant-head (positive) boundaries and leakage may produce.

Figure 8 illustrates a variation we have observed in recovery data. The recovery data for ideal conditions should approach $s/Q_n = 0$ as the $\log_{10}$ of the adjusted dimensionless time goes to zero. In some cases the curve is displaced upward (a in figure); in others it is displaced downward (b in figure). Two reasons for these departures are:

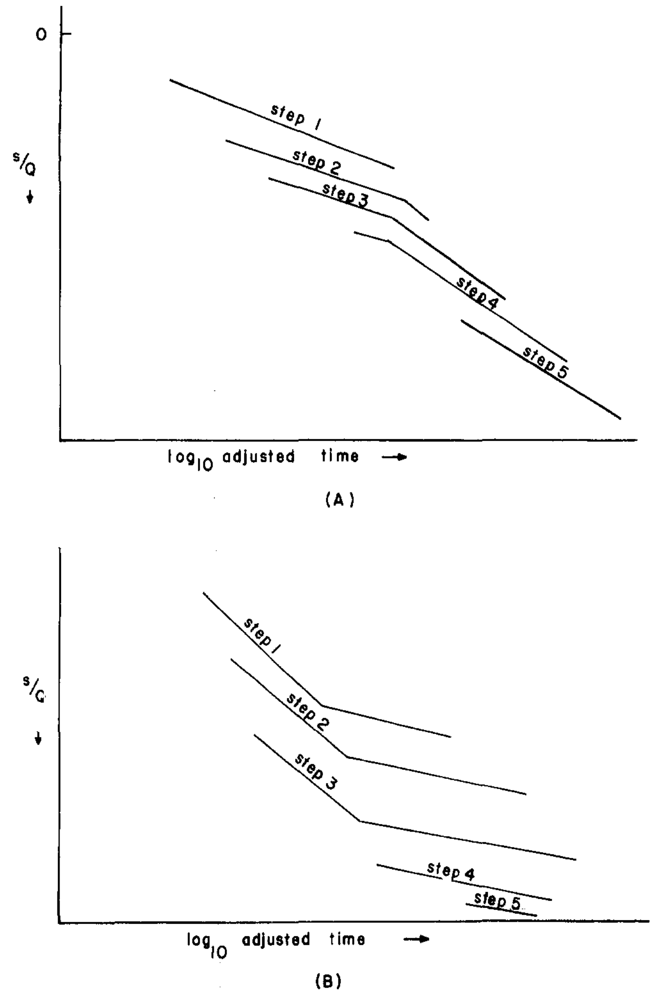

Fig. 7. (A) Pattern of adjusted time versus s/Q for pumping well near negative hydrologic boundaries. (B) Pattern of adjusted time versus s/Q for pumping well near positive hydrologic boundaries or in a leaky system.

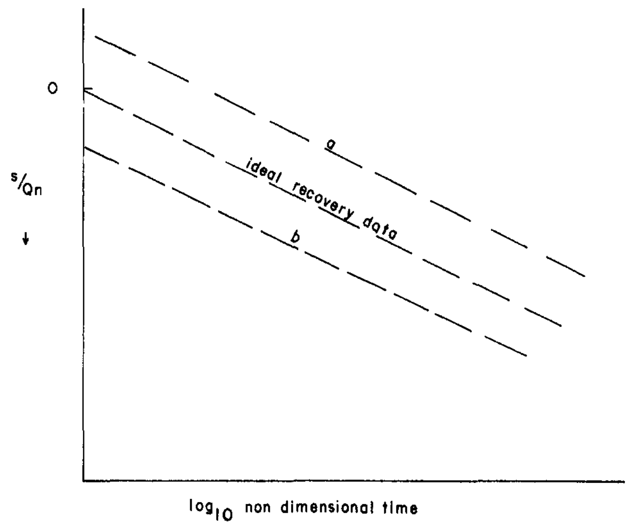

Fig. 8. Pattern of dimensionless time versus s/Q for pumping well where (a) water derives from sources other than storage and (b) more water discharged from storage than discharged from well or dewatering occurred.

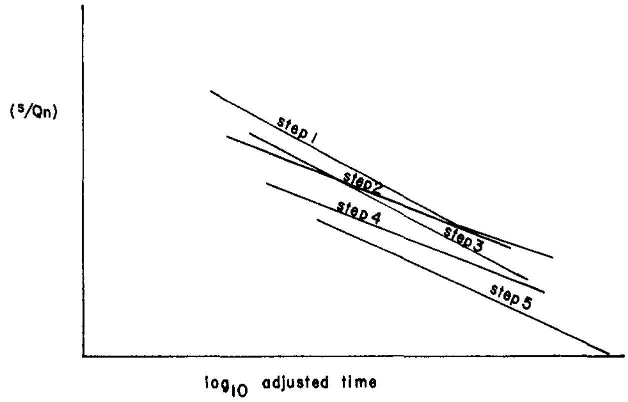

Fig. 9. Pattern of adjusted time versus s/Q where discharge or drawdown-measurements are not accurate.

(1) The drawdown and recovery data were not corrected adequately for precedent water-level trends, and

(2) The assumption that all water came from storage instantaneously may be violated.

Curve a of Figure 8 will come about if local recharge occurs; whereas, curve b will come about if water from storage is discharging nearby or if dewatering occurs.

Figure 9 illustrates the effect of poor data. The lines are straight but not parallel. This pattern reflects inaccuracies in the measurements taken during the test. The arithmetic average of the slopes of the straight lines should be used to estimate T. C and y cannot be obtained from these tests.

EXAMPLES

(1) By using the program in Appendix A, we analyzed a pumping test for a well near Angel Fire, New Mexico. The first and second columns of Table 1 contain the field data for this test and the other columns show each step of the calculations. The last column is the Jacob (1950) correction for unconfined aquifers. Figure 10 illustrates corrected s/Q_n (Jacob correction) versus adjusted time. The transmissivity obtained with the Jacob correction is 1160 gpd/ft.

By assuming that y equals one and using only steps 3, 4, and 5 with Equation (12) we obtained C to be approximately 1.8×10^{-3} ft/gpm^2. Since the second step produces an unrealistically small value for C, it was not included in the calculations. This may be an indication of unstable conditions near the well during the early part of the test.

(2) Figure 11 is a typical example of data from a test of a well near a hydrologic discontinuity.

(3) Figure 12 is a typical example of data obtained where the discharge data lack the necessary precision (or the discharge during each step may not have been sufficiently constant). The average slope probably gives a fair estimate of transmissivity, but no conclusions can be made about well efficiency.

(4) Figure 13 is a typical example of data obtained from a developing well. As such, they provide a basis for determining whether development should be continued.

Data Interpretation

Ideally, the rocks tapped by a pumping well have a unique transmissivity so where basic Theissian assumptions are satisfied, to determine transmissivity we need only determine the slope of a straight line obtained by plotting drawdown or recovery data on semilog paper. We integrate the data to obtain one slope. Our procedure is as follows:

(1) We plot the drawdown and recovery for the pumping well to the same scale on a sheet of semilog paper. If we have observation wells, we plot to the same scale as the pumping well the data for each observation well on its own sheet

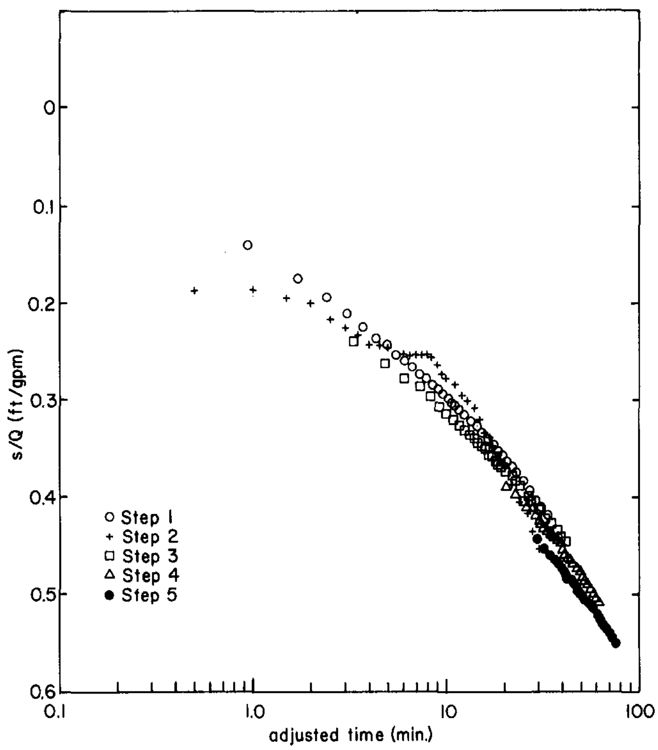

Fig. 10. Typical example of adjusted time versus s/Q used to compute transmissivity and well-loss coefficients.

Table 1. Pumping-Test Data

STEP 2

Q_1 = 2.3 gpm Q_2 = 14.7 gpm
τ_1 = 0.0 min τ_1' = 30.0 min
Length of screen (D) 200 feet

Time since pumping began (minutes)	Observed drawdown (feet)	β(t) Adjust-ment factor	Adjusted time (minutes)	s/Q_2 (ft/gpm)	$(s-s^2/2D)/Q_2$ (Jacob correction) (ft/gpm)
30.	1.04			.071	
30.5	2.05	1.90	0.95	.131	.139
31.0	2.53	1.71	1.71	.172	.171
31.5	2.85	1.61	2.42	.194	.192
32.0	3.09	1.54	3.09	.210	.209
32.5	3.29	1.49	3.73	.224	.222
33.0	3.45	1.46	4.37	.235	.233
33.5	3.58	1.42	4.98	.224	.241
34.0	3.71	1.40	5.59	.252	.250
34.5	3.81	1.38	6.19	.259	.257
35.0	3.91	1.36	6.78	.266	.263
35.5	4.01	1.34	7.36	.273	.270
36.0	4.10	1.32	7.94	.279	.276
36.5	4.18	1.31	8.51	.284	.281
37.0	4.25	1.30	9.08	.289	.286
37.5	4.32	1.29	9.65	.294	.291
38.0	4.39	1.28	10.21	.299	.295
38.5	4.46	1.27	10.77	.303	.300
39.0	4.51	1.26	11.32	.307	.303
39.5	4.56	1.25	11.87	.310	.307
40.0	4.63	1.24	12.42	.315	.311
41.0	4.72	1.23	13.51	.321	.317
42.0	4.82	1.22	14.60	.328	.324
43.0	4.92	1.21	15.68	.335	.331
44.0	5.01	1.20	16.75	.341	.337
45.0	5.11	1.19	17.81	.348	.343
46.0	5.19	1.18	18.87	.353	.348
47.0	5.27	1.17	19.93	.359	.354
48.0	5.35	1.17	20.99	.364	.359
49.0	5.43	1.16	22.04	.369	.364
50.0	5.51	1.15	23.08	.375	.370
52.0	5.64	1.14	25.17	.384	.372
54.0	5.79	1.14	27.25	.394	.388
56.0	5.91	1.13	29.32	.402	.396
58.0	6.04	1.12	31.38	.411	.405

STEP 3

Q_1 = 2.3 gpm, Q_2 = 14.7 gpm, Q_3 = 32.3 gpm
τ_1 = 0.0 min, τ_1' = τ_2 = 30.0 min, τ_2' = τ_3 = 60. min

Time since pumping began (minutes)	Observed drawdown (feet)	β(t) Adjust-ment factor	Adjusted time (minutes)	s/Q_3 (ft/gpm)	$(s-s^2/2D)/Q_3$ (Jacob correction) (ft/gpm)
60	6.16			.191	.188
60.5	7.66	6.82	3.41	.237	.233
61.0	8.40	5.01	5.01	.260	.255
61.5	8.93	4.19	6.29	.276	.270
62.0	9.30	3.70	7.40	.288	.281
62.5	9.62	3.37	8.42	.298	.291
63.0	9.90	3.12	9.36	.307	.299
63.5	10.16	2.93	10.24	.315	.307
64.0	10.35	2.77	11.08	.312	.312
64.5	10.53	2.64	11.89	.326	.317
65.0	10.70	2.53	12.67	.331	.322
65.5	10.86	2.44	13.42	.336	.327
66.0	10.99	2.36	14.16	.340	.331
66.5	11.11	2.29	14.88	.344	.334
67.0	11.23	2.23	15.58	.348	.338
67.5	11.35	2.17	16.27	.351	.341
68.0	11.47	2.12	16.95	.355	.345
68.5	11.57	2.07	17.61	.358	.348
69.0	11.68	2.03	18.27	.362	.351
69.5	11.78	1.99	18.92	.365	.354
70.0	11.88	1.96	19.56	.368	.357
71.0	12.07	1.89	20.82	.374	.362
72.0	12.24	1.84	22.05	.379	.367
73.0	12.41	1.79	23.27	.384	.372
74.0	12.55	1.75	24.46	.389	.376
75.0	13.13	1.71	25.65	.407	.393
76.0	12.86	1.68	26.81	.398	.385
77.0	13.00	1.65	27.97	.402	.389
78.0	13.13	1.62	29.12	.407	.393
79.0	13.26	1.59	30.25	.411	.397
80.0	13.39	1.57	31.38	.415	.401
82.0	13.62	1.53	33.61	.422	.407
84.0	13.84	1.49	35.82	.428	.414
86.0	14.02	1.46	38.01	.434	.419
88.0	14.22	1.43	40.18	.440	.425
90.0	14.41	1.41	42.33	.446	.430

STEP 4

Q_1 = 2.3 gpm, Q_2 = 14.7 gpm, Q_3 = 32.3 gpm and Q_4 = 38.5 gpm
τ_1 = 0.0 min, τ_1' = τ_2 = 30.0 min, τ_2' = τ_3 = 60 min and τ_4 = 90 min

Time since pumping began (minutes)	Observed drawdown (feet)	β(t) Adjust-ment factor	Adjusted time (minutes)	s/Q_4 (ft/gpm)	$(s-s^2/2D)/Q_4$ (Jacob correction) (ft/gpm)
90	14.41		--	0.374	0.361
90.5	14.99	41.86	20.93	0.389	0.375
91	15.36	23.65	23.65	0.399	0.384
91.5	15.61	17.00	25.51	0.405	0.390
92	15.84	13.49	26.99	0.411	0.395
92.5	16.04	11.30	28.26	0.417	0.400
93	16.17	9.80	29.39	0.420	0.403
93.5	16.31	8.69	30.42	0.424	0.405
94	16.45	7.85	31.38	0.427	0.410
94.5	16.57	7.17	32.39	0.430	0.413
95	16.67	6.63	33.15	0.433	0.415
95.5	16.77	6.18	33.98	0.436	0.417
96	16.86	5.80	34.77	0.438	0.419
96.5	16.95	5.47	35.54	0.440	0.422
97	17.03	5.18	36.29	0.442	0.424
97.5	17.12	4.94	37.02	0.445	0.426
98	17.19	4.72	37.74	0.446	0.427
98.5	17.27	4.52	38.44	0.449	0.429
99	17.41	4.35	39.13	0.452	0.433
99.5	17.44	4.19	39.81	0.453	0.433
100	17.51	4.05	40.47	0.455	0.435
101	17.64	3.80	41.78	0.458	0.436
102	17.78	3.59	43.06	0.462	0.441
103	17.90	3.41	44.31	0.465	0.444
104	18.03	3.25	45.54	0.468	0.447
105	18.13	3.12	46.75	0.471	0.450
106	18.24	3.00	47.95	0.474	0.452
107	18.33	2.89	49.13	0.476	0.454
108	18.41	2.79	50.30	0.478	0.456
109	18.56	3.12	51.46	0.482	0.460
110	18.65e	2.63	52.61	0.484	.0462
112	18.84e	2.49	54.89	0.484	0.466
114	19.03e	2.38	57.13	0.494	0.471
116	19.22	2.28	59.35	0.499	0.475
118	19.39	2.20	61.55	0.504	0.479
120	19.56	2.12	63.73	0.508	0.483

STEP 5

Q_1 =2.3 gpm, Q_2 =14.7 gpm, Q_3 =32.3 gpm, Q_4 =38.5 gpm and Q_5 =46.0 gpm.
τ_1 =0.0 min, τ_1' =τ_2 =30.0 min, τ_2' =τ_3 =60 min, τ_3' =τ_4 =90 min and τ_4' =τ_5

Time since pumping began (minutes)	Observed drawdown (feet	β(t) Adjust-ment factor	Adjusted time (minutes)	s/Q_5 (ft/gpm)	$(s-s^2/2D)/Q_5$ (Jacob correction) (ft/gpm)
120	19.56			0.425	0.404
120.5	20.42	58.24	29.12	0.444	0.421
121	20.86	32.83	32.83	0.453	0.430
121.5	21.16	18.75	35.32	0.460	0.436
122	21.39	14.68	37.28	0.465	0.440
122.5	21.60	15.57	38.92	0.470	0.444
123	21.80	13.46	40.37	0.474	0.448
123.5	21.96	11.91	41.67	0.477	0.451
124	22.09	10.72	42.87	0.480	0.454
124.5	22.23	9.78	43.99	0.483	0.456
125	22.35	9.01	45.05	0.486	0.459
125.5	22.47	8.37	46.05	0.488	0.461
126	22.49	7.83	47.01	0.489	0.461
126.5	22.68	7.37	47.93	0.493	0.465
127	22.77	6.97	48.82	0.495	0.467
127.5	22.87	6.62	49.68	0.497	0.469
128	22.96	6.31	50.52	0.499	0.470
128.5	23.03	6.04	51.33	0.501	0.472
129	23.12	5.79	52.13	0.503	0.474
129.5	23.19	5.51	52.91	0.504	0.475
130	23.28	5.37	53.67	0.506	0.477
131	23.42	5.01	53.16	0.509	0.479
132	23.58	4.72	56.60	0.511	0.481
133	23.58	4.46	58.01	0.513	0.482
134	23.68	4.24	59.38	0.515	0.484
135	23.78	4.05	60.72	0.517	0.486
136	23.96	3.88	62.04	0.521	0.490
137	24.11	3.73	63.34	0.524	0.493
138	24.23	3.59	64.61	0.527	0.495
139	24.35	3.47	65.87	0.529	0.497
140	24.46	3.36	67.12	0.532	0.499
142	24.69	3.16	69.57	0.537	0.504
144	24.91	3.00	71.97	0.542	0.508
146	25.11	2.86	74.34	0.546	0.512
148	25.32	2.74	76.67	0.550	0.516
150	25.51	2.63	78.97	0.555	0.519

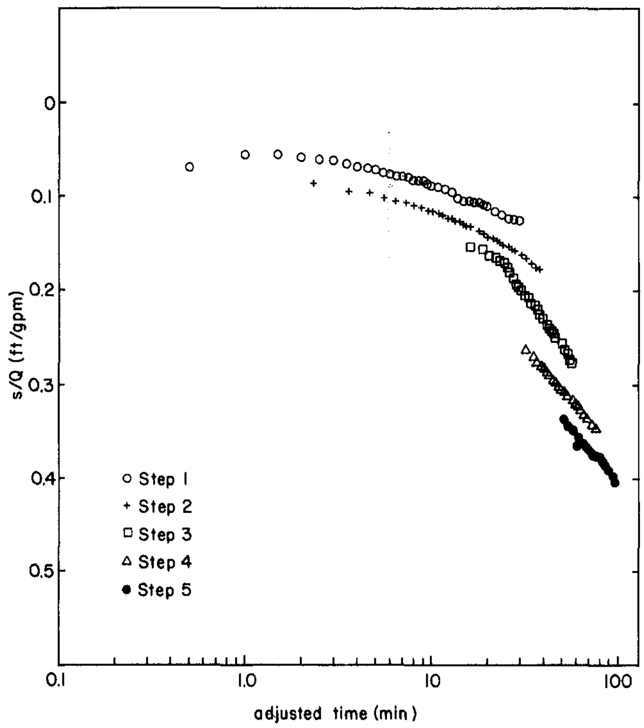

Fig. 11. Example of adjusted time versus s/Q for pumping well near hydrologic discontinuity.

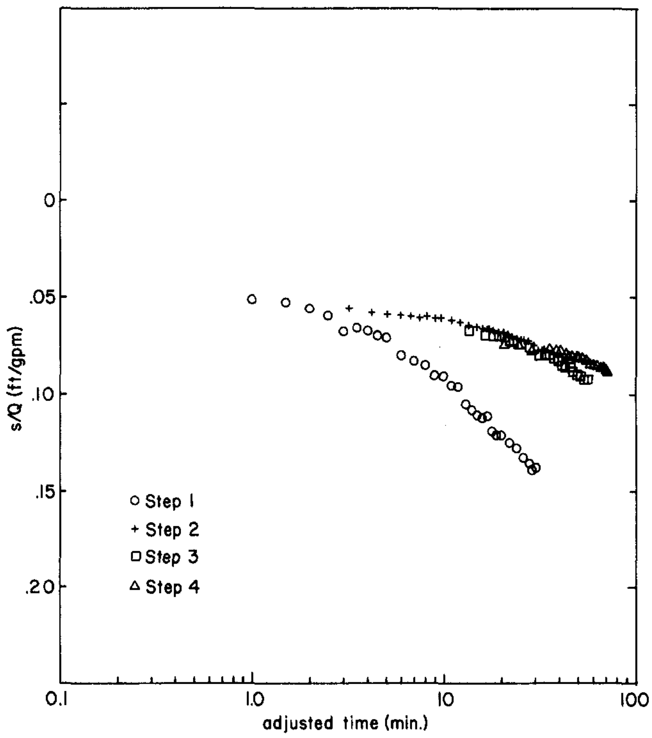

Fig. 13. Typical example of adjusted time versus s/Q for developing well.

of paper. (Clearly, we could plot all the data on one sheet of paper.)

(2) Next, we superimpose the semilog plots on a light table.

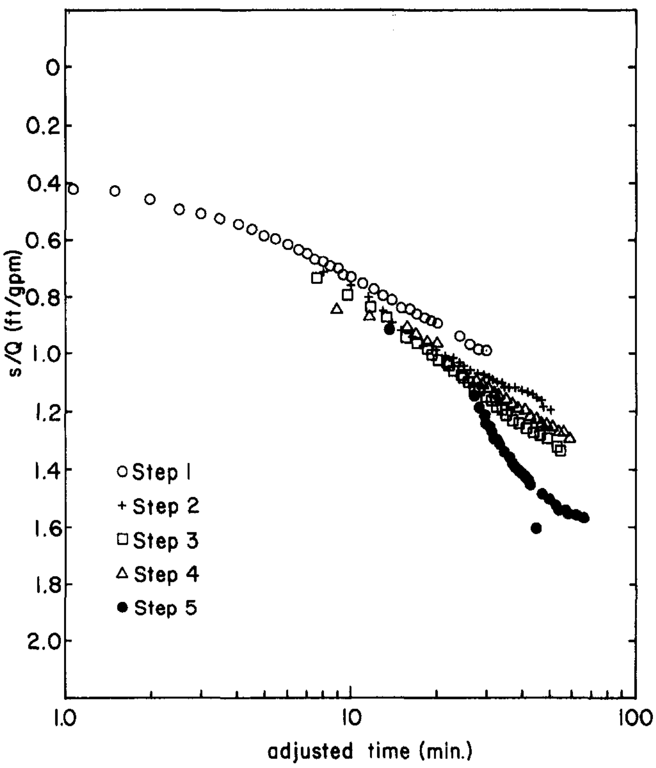

Fig. 12. Typical example of adjusted time versus s/Q for dubious data.

(3) If the data are ideal, the determination of the unique slope can be made easily with a pair of plastic triangles.

(4) If the data depart from the ideal, then no distinctive or unique slope becomes evident. We then look at the data pattern to see if it accounts for the departure.

Comparison of Short Tests with 24-Hour Tests

Table 2 compares the transmissivity and storativity determined from four step tests with those obtained from conventional 24-hour tests of the same wells. The rocks tapped by the well are fractured sandstones, siltstones, and shales. The wells are of different depths, have differing lengths of screen or perforated casing, and penetrate different thicknesses of saturated rock.

Country Club 3A Well is 200 feet deep, taps 150 feet of saturated rocks, and has 30 feet of screen in the bottom 50 feet. The agreement between T's is good. The storativity of the step test probably reflects elastic-storage components only [specific storage (S_S) of 5×10^{-6}/ft]; whereas, the storativity obtained from the 24-hour test probably includes a component of water derived from porosity storage.

The Peralta Well is in a fault zone. The step-test data characterize the rocks near the well. The

Table 2. Comparison of Transmissivity and Storativity Obtained by Step Tests and 24-Hour Tests

Well	*Transmissivity (gpd/ft)* Step test	24-hour test	
Country Club 3A	2900	1800- 3900	
Peralta	88000	very large	early data
		6900-13200	late data
Coffey 2	18200	7500-45000	early data
		4500- 5600	late data
Base Lodge 1	14700	8300-11000	
	Storativity Step test	24-hour test	
Country Club 3A	6.5×10^{-4}	$3.9 \times 10^{-3} - 9.1 \times 10^{-3}$	
Coffey 2	$.52 \times 10^{-4} - 8.8 \times 10^{-4}$	$52 \times 10^{-4} - .16 \times 10^{-4}$	early data
		$1.6 \times 10^{-2} - 8.8 \times 10^{-2}$	late data

early data from the 24-hour test suggest a very large transmissivity; whereas, the later data (after 500 minutes) probably reflect an "average" property of the rocks.

The Coffey 2 Well is in a complex hydrogeologic setting. It is in a fault zone near a stream. Conventional analytical techniques (Theis and Jacob) fail. The step-test data give transmissivities that agree well with the early 24-hour data and give storativities that suggest that only an elastic storage component operated $[S_s \sim 10^{-7}/\text{ft}]$. The late 24-hour data suggest that delayed yield or slow drainage occurs.

The transmissivity obtained from the Base-Lodge Well step-test data is the same order of magnitude as those obtained from the 24-hour test data.

These comparisons warrant the following conclusions:

(1) In rocks of diverse hydraulic conductivity, step tests measure the transmissivity and elastic storage characteristics of the rocks near the well reliably.

(2) In regions where the rocks are essentially isotropic and homogeneous, short step tests measure the transmissivic characteristics of the rocks as reliably as 24-hour tests.

(3) In regions where the rocks are neither isotropic nor homogeneous, longer tests give more realistic average values for transmissivity.

(4) Long tests tend to be more difficult to interpret in complex geohydrologic settings.

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APPENDIX A

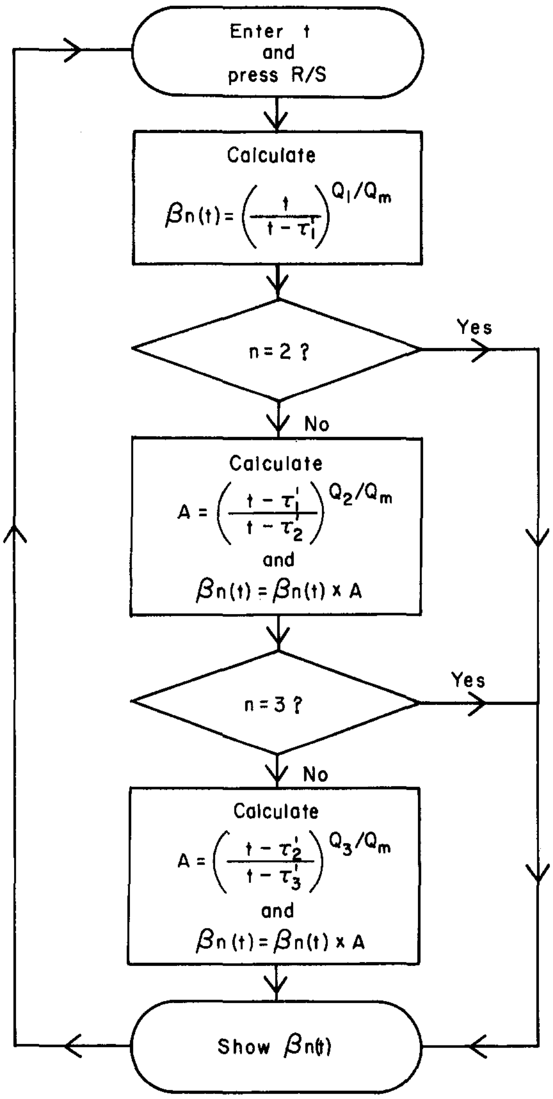
1. Storage and program

R_0	R_1	R_2	R_3	R_4	R_5	R_6	R_7
t	τ_1'	τ_2'	τ_3'	Q_1/Q_m	Q_2/Q_m	Q_3/Q_m	m

1	STO	0	23	fy^x	
2	↑		24	x	
3	↑		25	3	
4	RCL	1	26	RCL	7
5	−		27	−	
6	÷		28	gx=o	
7	RCL	4	29	GTO	42
8	fy^x		30	⇄	
9	2		31	RCL	0
10	RCL	7	32	RCL	2
11	−		33	−	
12	gx=o		34	RCL	0
13	GTO	42	35	RCL	3
14	⇄		36	−	
15	RCL	0	37	÷	
16	RCL	1	38	RCL	6
17	−		39	fy^x	
18	RCL	0	40	x	
19	RCL	2	41	GTO	00
20	−		42	⇄	
21	÷		43	GTO	00
22	RCL	5			

If m=2 R_5 and R_6 and if m=3 R_6 can be left 0

2. Flow Chart



3. Operating instructions.
   A. Load the program then press fPRGM.
   B. Load necessary storage locations.
   C. Load the time on X register and press R/S.
   D. Repeat C for a given step as much as necessary.
   E. For a different pumping step restart from item B.

4. Examples.
   1. Calculation of adjusted times of the second step (see Table 1).
      $R_1 = 30.0$
      $R_4 = Q_1/Q_2 = 2.3/14.7$
      $R_7 = 2$
   and we obtain
      $\beta_2(38.0) = 1.28$
   2. Calculation of adjusted time of the third step
      $R_1 = 30.$ $R_2 = 60.$
      $R_4 = Q_1/Q_3 = 2.3/32.3$ $R_5 = Q_2/Q_3 = 14.7/32.3$
      $R_7 = 3$
   and we obtain
      $\beta_3(65.5) = 2.44$
   3. Calculation of adjusted time of the fourth step.
      $R_1 = 30.$ $R_2 = 60.$ $R_3 = 90.$
      $R_4 = 2.3/38.5$ $R_5 = 14.7/48.5$ $R_6 = 32.3/38.5$
      $R_7 \neq 2$ or 3
      $\beta_4(95.0) = 6.63$ and so on.